

# Christopher Musco

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## Academic Positions

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| <b>New York University</b><br><i>Assistant Professor, Tandon School of Engineering</i><br><i>Department of Computer Science and Engineering</i> | <b>New York, NY</b><br><i>2019 - present</i> |
| <b>Microsoft Research</b><br><i>Visiting Researcher, Machine Learning and Optimization Group</i>                                                | <b>Redmond, WA</b><br><i>Summer 2020</i>     |
| <b>Princeton University</b><br><i>Research Instructor, Department of Computer Science</i>                                                       | <b>Princeton, NJ</b><br><i>2018 – 2019</i>   |
| <b>Massachusetts Institute of Technology</b><br><i>Research and Teaching Assistant, Theory of Computation Group</i>                             | <b>Cambridge, MA</b><br><i>2013 – 2018</i>   |
| <b>Yahoo Labs</b><br><i>Research Intern, Scalable Machine Learning Group</i>                                                                    | <b>New York, NY</b><br><i>Summer 2015</i>    |

## Research Areas

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foundations of data science • scalable machine learning • numerical linear algebra • theory of algorithms  
randomization • dimensionality reduction • sketching & streaming • spectral graph algorithms • optimization

## Education

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| <b>Massachusetts Institute of Technology</b><br><i>Ph.D. in Computer Science</i><br>Advisor: Jonathan Kelner<br>Thesis: Faster Linear Algebra for Data Analysis and Machine Learning<br>M.S. Thesis: Dimensionality Reduction for Sparse and Structured Matrices | <b>Cambridge, MA</b><br><i>2013 – 2018</i> |
| <b>Yale University</b><br><i>B.S. in Applied Mathematics, B.S. in Computer Science</i>                                                                                                                                                                           | <b>New Haven, CT</b><br><i>2008 – 2012</i> |

## Publications

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*Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science. Any exceptions are marked with a \*, which indicates that authors are ordered by contribution.*

**Chemically-informed data-driven optimization (ChIDDO): Leveraging physical models and Bayesian learning to accelerate chemical research\***. Daniel Frey, Juhee Shin, Christopher Musco, Miguel A. Modestino. *Reaction Chemistry & Engineering* 2022.

**Streaming Approach to In Situ Selection of Key Time Steps for Time-Varying Volume Data\***. Mengxi Wu, Yi-Jen Chiang, Christopher Musco. *EuroVis* 2022.

**Low-memory Krylov subspace methods for optimal rational matrix function approximation**. Tyler Chen, Anne Greenbaum, Cameron Musco, Christopher Musco. *Preprint* 2021.

**Fast Regression for Structured Inputs.** Raphael Meyer, Cameron Musco, Christopher Musco, David P. Woodruff Sidford, Samson Zhou. *International Conference on Learning Representations (ICLR)* 2022.

**Fast and Near-Optimal Diagonal Preconditioning.** Arun Jambulapati, Jerry Li, Christopher Musco, Aaron Sidford, Kevin Tian. *Preprint* 2021.

**Active Sampling for Linear Regression Beyond the  $\ell_2$  Norm.** Cameron Musco, Christopher Musco, David P. Woodruff, Taisuke Yasuda. *Preprint* 2021.

**How to Quantify Polarization in Models of Opinion Dynamics.** Christopher Musco, Indu Ramesh, Johan Ugander, R. Teal Witter *Preprint* 2021.

**Error Bounds for Lanczos-based Matrix Function Approximation.** Tyler Chen, Anne Greenbaum, Cameron Musco, Christopher Musco. *SIAM Journal on Matrix Analysis and Applications (SIMAX)* 2021.

**Linear and Sublinear Time Spectral Density Estimation.** Vladimir Braverman, Aditya Krishnan, Christopher Musco. *ACM Symposium on Theory of Computing (STOC)* 2022.

**Dynamic Trace Estimation.** Prathamesh Dharangutte, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2021.

**Finding the Mode of a Kernel Density Estimate.** Jasper C.H. Lee, Jerry Li, Christopher Musco, Jeff M. Phillips, Wai Ming Tai. *European Symposium on Algorithms (ESA)* 2021.

**Correlation Sketches for Approximate Join-Correlation Queries\*.** Aécio Santos, Aline Bessa, Fernando Chirigati, Christopher Musco, Juliana Freire. *ACM Special Interest Group on Management of Data Conference (SIGMOD)* 2021.

**Public Transport Planning: When Transit Network Connectivity Meets Commuting Demand\*.** Sheng Wang, Yuan Sun, Christopher Musco, Zhifeng Bao. *ACM Special Interest Group on Management of Data Conference (SIGMOD)* 2021.

**Simple Heuristics Yield Provable Algorithms for Masked Low Rank Approximation.** Cameron Musco, Christopher Musco, David Woodruff. *Innovations in Theoretical Computer Science (ITCS)* 2021.

**Hutch++: Optimal Stochastic Trace Estimation.** Raphael Meyer, Cameron Musco, Christopher Musco, David P. Woodruff. *SIAM Symposium on Simplicity in Algorithms (SOSA)* 2021.

**Graph Learning for Inverse Landscape Genetics.** Prathamesh Dharangutte, Christopher Musco. *AAAI Conference on Artificial Intelligence (AAAI)* 2021. Short version in NeurIPS 2020 AI for Earth Sciences workshop.

**Fourier Sparse Leverage Scores and Approximate Kernel Learning.** Tamás Erdélyi, Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2020. Invited for spotlight presentation (4% of submissions).

**Near Optimal Linear Algebra in the Online and Sliding Window Models.** Vladimir Braverman, Petros Drineas, Cameron Musco, Christopher Musco, Jalaj Upadhyay, David P. Woodruff, Samson Zhou. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2020.

**The Statistical Cost of Robust Kernel Hyperparameter Turning.** Raphael A. Meyer, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2020.

**Low-Rank Toeplitz Matrix Estimation via Random Ultra-Sparse Rulers.** Hannah Lawrence, Jerry Li, Cameron Musco, Christopher Musco. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. 2020.

**Sample Efficient Toeplitz Covariance Estimation.** Yonina Eldar, Jerry Li, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

**Fast and Space Efficient Spectral Sparsification in Dynamic Streams.** Michael Kapralov, Aida Mousavifar, Cameron Musco, Christopher Musco, Navid Nouri, Aaron Sidford, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

**Understanding Filter Bubbles and Polarization in Social Networks.** Uthsav Chitra, Christopher Musco. *ACM International Conference on Web Search and Data Mining (WSDM) 2020*. Preliminary version in KDD Workshop on Issues of Sentiment Discovery and Opinion Mining.

**A Universal Sampling Method for Reconstructing Signals with Simple Fourier Transforms.** Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *ACM Symposium on Theory of Computing (STOC) 2019*.

**Learning Networks from Random Walk-Based Node Similarities.** Jeremy Hoskins, Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *Conference on Neural Information Processing Systems (NeurIPS) 2018*.

**Eigenvector Computation and Community Detection in Asynchronous Gossip Models.** Frederik Mallmann-Trenn, Cameron Musco, Christopher Musco. *International Colloquium on Automata, Languages, and Programming (ICALP) 2018*.

**Minimizing Polarization and Disagreement in Social Networks.** Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *The Web Conference (WWW) 2018*.

**Stability of the Lanczos Method for Matrix Function Approximation.** Cameron Musco, Christopher Musco, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA) 2018*.

**Recursive Sampling for the Nyström Method.** Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS) 2017*.

**Random Fourier Features for Kernel Ridge Regression: Approximation Bounds and Statistical Guarantees.** Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *International Conference on Machine Learning (ICML) 2017*.

**Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling.** Michael B. Cohen, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA) 2017*.

**Determining Tournament Payout Structures for Daily Fantasy Sports.** Christopher Musco, Maxim Sviridenko, Justin Thaler. *SIAM Algorithm Engineering and Experiments (ALENEX) 2017*.

**Principal Component Projection Without Principal Component Analysis.** Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *International Conference on Machine Learning (ICML) 2016*.

**Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition.** Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS) 2015*. Invited for full oral presentation (1 of 15, < 1% of submissions). Also presented at the 2016 Copper Mountain Conference on Iterative Methods

**Dimensionality Reduction for K-Means Clustering and Low Rank Approximation.** Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *ACM Symposium on Theory of Computing (STOC) 2015*.

**Principled Sampling for Anomaly Detection.** Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiroglou-Douskos, Martin Rinard. *Network and Distributed System Security Symposium (NDSS) 2015*.

**Uniform Sampling for Matrix Approximation.** Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *Innovations in Theoretical Computer Science (ITCS) 2015*.

**Single Pass Spectral Sparsification in Dynamic Streams.** Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *IEEE Symposium on Foundations of Computer Science (FOCS) 2014*, Special Issue SIAM Journal on Computing 2017.

## Funding

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### NSF CAREER Award

*Frontiers in Matrix Sketching.*

PI, \$562,224

2021-2026

**DOE Office of Science Grant***Leverage Score Sampling for Parametric PDEs.**2021-2023*

PI (w/ Chinmay Hegde, NYU), \$300,000

**NSF III Medium***Dataset Search and Ranking for Data Augmentation and Explanation.**2021-2024*

Co-PI (w/ Juliana Freire, NYU), \$1,093,195

**Current Graduate Student and Postdoc Advising**

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Tyler Chen. Postdoc, New York University.

*Starting 2022*

Majid Daliri. Ph.D., New York University.

*Starting 2022*

Aarshvi Gajjar. Ph.D., New York University (with Prof. Chinmay Hedge).

*2021 - present*

Apoorv Singh. Ph.D., New York University.

*2020 - present*

R. Teal Witter. Ph.D., New York University (with Prof. Lisa Hellerstein).

*2020 - present*

Raphael Meyer. Ph.D., New York University.

*2019 - present*

Indu Ramesh. M.S., New York University.

*2020 - present*

Xinyu Luo. M.S., New York University.

*2020 - present***Past Advising**

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Aline Bessa. Postdoc, New York University.

*2021*

Prathamesh Dharangutte. M.S., New York University.

*2019 - 2021*

John Shin. M.S., New York University.

*2019 - 2020*

I have advised over a dozen undergraduate guided studies and research projects.

**Ph.D. Thesis Committees**

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Francis Williams, NYU

*Defended 2021.*

Yiming Zhang, NYU

*Defending 2022.***Teaching**

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**Machine Learning, Graduate (NYU CS-GY 6923)***Course Instructor**Spring 2022***Algorithmic Machine Learning and Data Science (NYU CS-GY 6763)***Course Instructor**Fall 2019, 2020, 2021***Introduction to Machine Learning (NYU CS-UY 4563)***Course Instructor**Spring 2020***Advanced Algorithm Design (Princeton COS 521)***Course Instructor**Fall 2018***Advanced Algorithms (MIT 6.854/18.415)***Teaching Assistant**Spring 2016***Technical Communication Skills for Graduate Students (MIT 6.S977)***Workshop Leader**Spring 2016*

## Mentorship and Outreach

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### **SIAM/MathWorks Math Modeling Challenge**

*Problem Development Committee, Director of Technical Computing Judging* 2018 – present

Support SIAM's annual math modeling competition, which gives high school students an opportunity to apply mathematics, statistics, and data science to a real-world problem. Serve on committee in charge of writing practice and contest problems. Lead judging and award presentation for student Technical Computing prizes.

### **MIT Graduate Communication Lab**

*Communication Adviser* 2015 – 2018

Helped to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offered feedback, guidance, and coaching to individual students during weekly office hours.

### **CSAIL Algorithms Office Hours**

*Member* 2016 – 2018

Advised researchers, generally in applied fields, on framing and solving algorithmic problems arising in their work.

### **Research Science Institute at MIT**

*Summer Research Adviser* 2014

Primary supervisor for a high school student participating in the prestigious RSI summer research program. Advisee's project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

## Talks and Presentations

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### **Sublinear Time Spectral Density Estimation**

Oxford University, Computational Mathematics and Applications Seminar Feb. 2022

University of British Columbia, Applied Math Seminar Jan. 2022

Workshop on Algorithms for Large Data Aug. 2021

Cornell University, Scientific Computing and Numerics Seminar Mar. 2021

### **Optimal Stochastic Trace Estimation**

New York University, Numerical Analysis and Scientific Computing Seminar April 2022

University of Michigan, Randomized Numerical Linear Algebra Seminar Nov. 2021

Stanford University, Operations Research Seminar Oct. 2020

### **A TCS Perspective on Iterative Eigensolvers**

Complexity of Matrix Computations Seminar Oct. 2021

### **Mathematical Modeling and Coding in the Classroom**

Mu Alpha Theta National Convention July 2021

### **Randomized Algorithms for Probing the Spectrum of a Matrix**

George Mason University, AI Tea Series Apr. 2021

### **Importance Sampling for Approximating High-Dimensional Kernel Matrices**

Emory University, Numerical Analysis and Scientific Computing Seminar Feb. 2021

### **Structured Covariance Estimation**

Oxford Statistics and Machine Learning in Finance Seminar June 2022

Yahoo! Research, Seminar April 2020

Johns Hopkins University, Theory Seminar Oct. 2019

Symposium on Discrete Algorithms (SODA) Jan. 2020

### **From Randomized Linear Algebra to Randomized Functional Analysis**

SIAM Mathematics of Data Science Conference May 2020

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|-------------------------------------------------------------------------------------|-------------------|
| University of Utah, Data Science Seminar                                            | <i>Jan. 2020</i>  |
| Rutgers University, DIMACS RandNLA Workshop                                         | <i>Sept. 2019</i> |
| <b>Learning Signals with Simple Fourier transforms</b>                              |                   |
| Symposium on Theory of Computing (STOC)                                             | <i>June 2019</i>  |
| Columbia University, Theory Seminar                                                 | <i>Oct. 2018</i>  |
| UC Berkeley, Simons Institute Workshop                                              | <i>Sept. 2018</i> |
| Princeton University, Theory Seminar                                                | <i>Sept. 2018</i> |
| <b>The Lanczos Method in Data Science: Challenges and Opportunities</b>             |                   |
| SIAM Conference on Parallel Processing for Scientific Computing, Minisymposium      | <i>Feb. 2020</i>  |
| Georgetown University, CS Seminar                                                   | <i>Feb. 2018</i>  |
| Symposium on Discrete Algorithms (SODA)                                             | <i>Jan. 2018</i>  |
| <b>Unifying Random Features and Leverage Scores for Kernel Matrix Approximation</b> |                   |
| Institute for Advanced Study, Computer Science and Discrete Mathematics Seminar     | <i>Feb. 2018</i>  |
| Harvard University, guest lecture in "Sketching Algorithms for Big Data"            | <i>Nov. 2017</i>  |
| Microsoft Research, Redmond                                                         | <i>Sept. 2017</i> |
| <b>Sketching and Sampling Methods for Large Scale Linear Algebra</b>                |                   |
| Twitter, Cambridge                                                                  | <i>July 2017</i>  |
| <b>Recursive Sampling for the Nyström Method</b>                                    |                   |
| Neural Information Processing Systems (NeurIPS) poster session                      | <i>Dec. 2017</i>  |
| New England Machine Learning Day poster session                                     | <i>May 2017</i>   |
| <b>Algorithms for Determining Tournament Payout Structures</b>                      |                   |
| Meeting on Algorithm Engineering and Experiments (ALENEX)                           | <i>Jan. 2017</i>  |
| <b>Introduction to Linear Sketching</b>                                             |                   |
| MIT, Theory Retreat                                                                 | <i>Sept. 2016</i> |
| <b>Principal Component Projection without Principal Component Analysis</b>          |                   |
| International Conference on Machine Learning (ICML)                                 | <i>June 2016</i>  |
| <b>Ridge Leverage Score Sampling</b>                                                |                   |
| National Institute of Informatics, Shonan Meeting                                   | <i>July 2016</i>  |
| MIT, Algorithms & Complexity Seminar                                                | <i>July 2016</i>  |
| University of Utah, Data Group Meeting                                              | <i>Jan. 2016</i>  |
| <b>Randomized Block Krylov Methods</b>                                              |                   |
| Neural Information Processing Systems (NeurIPS) poster session                      | <i>Dec. 2015</i>  |
| <b>Large Scale Column Subset Selection</b>                                          |                   |
| Yahoo Labs, Science Week poster session                                             | <i>July 2015</i>  |
| <b>Dimensionality Reduction for k-Means Clustering</b>                              |                   |
| IBM T.J. Watson, Mathematical Sciences research seminar                             | <i>Aug. 2015</i>  |
| Symposium on Theory of Computing (STOC)                                             | <i>June 2015</i>  |
| MIT, Theory Lunch                                                                   | <i>June 2014</i>  |
| <b>Principled Sampling for Anomaly Detection</b>                                    |                   |
| MIT, Defense Advanced Research Projects Agency site visit                           | <i>Mar. 2015</i>  |
| Network and Distributed System Security Symposium (NDSS)                            | <i>Feb. 2015</i>  |

## Uniform Sampling for Matrix Approximation

Yahoo Labs, Scalable Machine Learning research seminar

June 2015

## Single Pass Spectral Sparsification in Dynamic Streams

MIT, Annual Sublinear Algorithms Day poster session

Apr. 2015

Harvard University, Theory Seminar

Nov. 2014

Foundations of Computer Science (FOCS)

Oct. 2014

## Service

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### Organization.....

NYU Computer Science Theory “Pandemic Presentations” Workshop

2022

SIAM Annual Meeting Minisymposium Organizer

2021

SIAM Mathematics of Data Science Minisymposium Organizer

2020

NYU Tandon Algorithmic Data Science Reading Group

2019 - present

### Department Service.....

Faculty Search Committee

2019, 2021, 2022

Online Faculty Oversight Committee

2021 - present

Ph.D. Admissions Committee

2019 - present

### Conference Programming.....

STOC Program Committee Member

2023

FOCS Program Committee Member

2022

NeurIPS Area Chair

2019, 2021, 2022

AAAI Senior Area Chair

2022

### Conference Reviewing.....

STOC, FOCS, SODA, ICALP, ITCS, ICML, NeurIPS, AAAI, AISTATS, SOCG, ESA, APPROX, RANDOM, STACS, IWOCA, WALCOM, NDSS

### Journal Reviewing.....

Journal of the ACM, SIAM Journal on Computing, ACM Transactions on Algorithms, Theoretical Computer Science, Journal of Machine Learning Research, SIAM Journal on Matrix Analysis and Applications, SIAM Journal on Scientific Computing, SIAM Journal on Mathematics of Data Science, Science Advances, Mathematical Programming, IEEE Transactions on Signal Processing, IEEE Transactions on Knowledge and Data Engineering, International Journal of Foundations of Computer Science, IEEE Journal on Selected Areas in Information Theory, BIT Numerical Mathematics

### Grants Panels.....

National Science Foundation 2019, 2022 (2)

### Notables.....

ICML 2018 Outstanding Reviewer, ICML Expert Review 2021

## Invited Workshops and Events

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### Rutgers, DIMACS

*Invited participant*

2019

Workshop on Randomized Numerical Linear Algebra, Statistics, and Optimization.

**UC Berkeley, Simons Institute for the Theory of Computing**

*Invited participant*

2018

Workshop on Randomized Numerical Linear Algebra and Applications.

**Japan National Institute of Informatics**

*Invited participant*

2016

Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra.

**Honors and Awards**

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**National Science Foundation**

2014-2017

*Graduate Research Fellowship, recipient.*

**Yale University**

2012

*Cum Laude, Distinction in Both Majors.*

**Other Professional Experience**

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**Redfin**

**Seattle, WA**

*Software Engineer, Data Team*

2012 – 2014

**Elysium Digital**

**Cambridge, MA**

*Summer Consultant*

2011

**Amicus**

**New Haven, CT**

*Software Developer*

2010 – 2011